

Department of Earth and Environmental Sciences

The Geology of the Mam Tor, Derbyshire



Introduction

The Mam Tor area - Winnats Pass (Fig 1) provides an excellent opportunity to examine a variety of geology and geological processes, both ancient and modern. The rocks of Carboniferous age provide a range of lithologies and fossils which formed in different environments - erosion levels are perfect for appreciating the palaeogeography of the area 360-330Ma. The area contains two different types of mineralisation (ancient lead + fluorite mineralisation, recent bog iron ore) and there is also evidence of hydrocarbon migration. The destructive Mam Tor landslip is both an example of a geological hazard and an opportunity to study some active faulting.

The main lithologies and their environment of deposition. The oldest rocks we will examine are the Lower Carboniferous (Dinantian) limestones (Fig 2). These limestones were formed in hot tropical seas into which little terrigenous material was flowing. The Peak District area south of Castleton formed a shallow carbonate platform (shelf) on which large amounts of limestone accumulated throughout the Dinantian (Fig 3a). The Castleton area, however, represents the very edge of that platform and to the north were deeper waters. Present erosion (Fig. 3d) has conveniently exposed the original slope between the shallow and deeper waters and gives us an impression of the original relationships between the rocks at the time of their formation.

The topographically high areas at the top of Winnats Pass and Treak Cliff were the top of the old marine slope and the edge of the platform, whereas Winnats Pass itself and the slope down into the Hope Valley are the original palaeoslope leading to the deeper basinal areas 130m below. The platform limestone to the south of the Castleton-Mam Tor area is the well-bedded Bee Low limestone. The rocks we shall examine are part of what has been termed the 'Apron Reef' that developed at the margin of the platform and on the palaeoslope. However, it was not quite a coral reef as we know them today because true reef building organisms had not been present, although it did form in similar conditions to present day reefs, such as found in the Bahamas today.

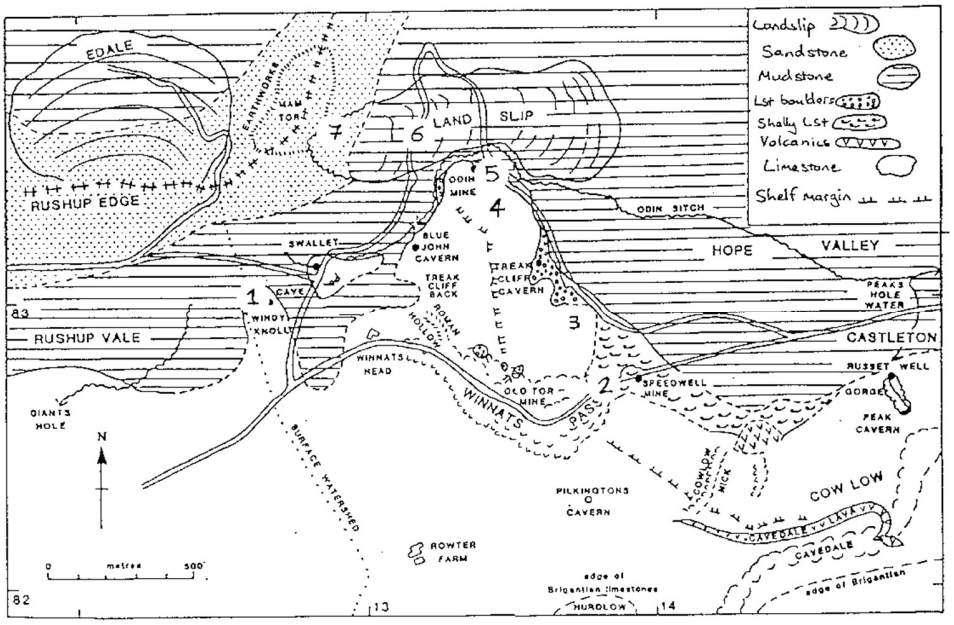


Figure 1. Geological sketch map of the Mam Tor – Winnats Pass area.

At the margin of the platform, carbonate sands accumulated (a grainstone) where limestone fragments were worked by wave action. We can see these at Windy Knoll where they have been termed ' back reef' limestones. On the slope, extensive biotas developed resulting in highly fossiliferous beds, giving the impression of a reef. In places algal stromatolite concentrations developed. These have been termed 'algal reefs'. At the base of the slope there are accumulations of brecciated limestone which formed from blocks that tumbled down the marine slope. The Carboniferous Limestone platform formed because it was relatively stable (buoyant) compared to the basinal areas. There were sea level fluctuations during the Dinantian and during a time of sea regression the so called 'beach beds' comprising very abundant brachiopod shells formed at the base of the palaeoslope. They can be seen opposite the entrance to Speedwell cavern.

The limestones are highly fossiliferous with an abundant fauna of corals (Fig. 4) *(Lithostrotion, Dibunophyllum),* brachiopods *(Productus, Pugnax, Schizophoria),* and goniatites *(Beyrichoceras).* The most common fossils are crinoids, although as we only see their stems, they are difficult to classify.

After the limestones came deposition of the early Upper Carboniferous (Namurian) mudstones and sandstones (Fig 3c) that progressively overlapped the carbonate platform (Fig. 5) and heralded the coming of deltas into the area. The deltas originated in the north, spreading southwards. At first there were deep water muds (the Edale Shales) which formed far from the delta front. As the deltas progressed, deltaic sands (the Mam Tor beds were deposited, representing the encroachment of the front apron of delta.

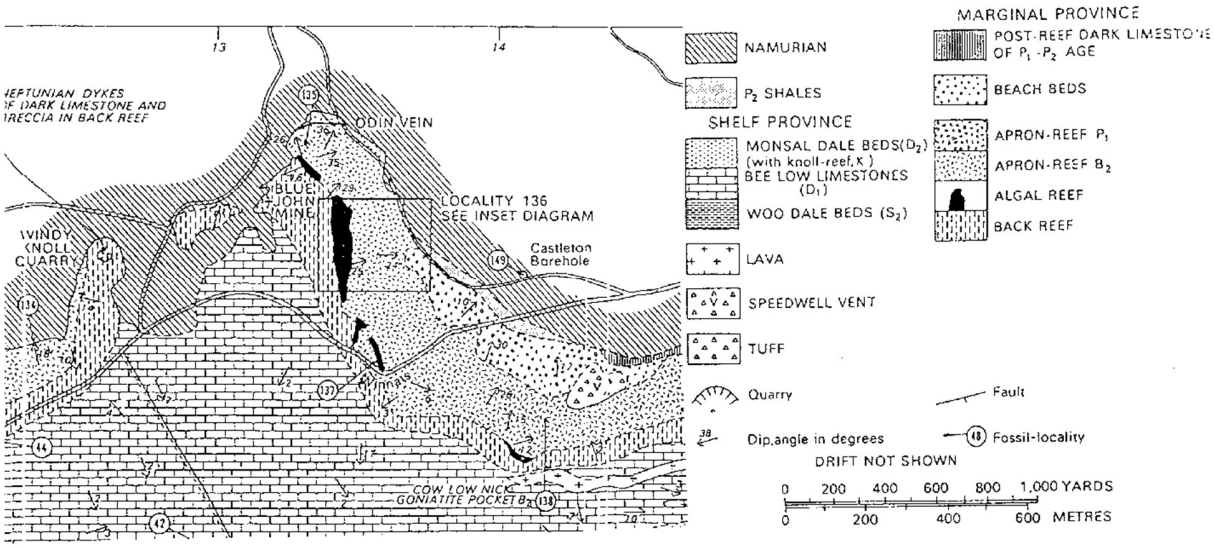


Figure 2. Limestones of the Treak Cliff area (Modified from the BGS Memoir 99).

Windy Knoll (SE 126 830)

In the small quarry at Windy Knoll we can examine the carbonate sands of the platform margin. The granular nature of the limestone can be seen, the grains being spherical, resembling ooids and formed in relatively turbulent waters. The rocks are also cut by fractures filled with brecciated dark limestone. These are 'neptunian dykes' which formed as cavities in the rock on the seafloor and became filled by later carbonate sediment and rock fragments

(Fig 3b). In addition the limestone contains brown, glutinous hydrocarbons 'Elaterite'. These are residues of degraded hydrocarbons that, in the past, migrated from their sources in basins surrounding the Carbonate platform, up the contact between the shales and limestones, using the same pathway as the ore fluids that caused the fluorite mineralisation. Evidence suggests the oil was produced in the late Carboniferous.

The cave seen at Windy Knoll was one of those excavated by Professor Boyd-Dawkins in the 19th century. It contained 5000 skeletons of mainly bison and reindeer. The locality was a swallet hole which accumulated the bones of animals using the migration route between Edale and the Cheshire Plain in the early post-glacial period.

Winnats Pass (SE 139 827)

This deeply incised feature (Fig. 1) is thought to have evolved in several stages, commencing as a channel cutting the original sediment shelf during the Lower Carboniferous. Its present day morphology is believed to be the result of erosion by melt-water run-off during the Pleistocene ice age (Fig. 3d).These melt-waters will have removed the original, softer shales deposited in the ancient vallev. The limestones of the pass are of two types a) fine-grained with sparse fossils and b) coarse grained limestone with abundant shell fragments. At the lower end of the pass, at the Speedwell cavern entrance we can examine the 'beach beds' (Fig 2) which comprise broken and worked shell fragments as well as darker, fine-grained limestones more typical of the deposition in the basinal areas. The shell fragments are mainly brachiopods that are typically found in the shallow water of the shelf and have been transported down to the base of the off the slope, and vertical joints in the rocks contain fluorite, evidence of mineralising fluid migration.

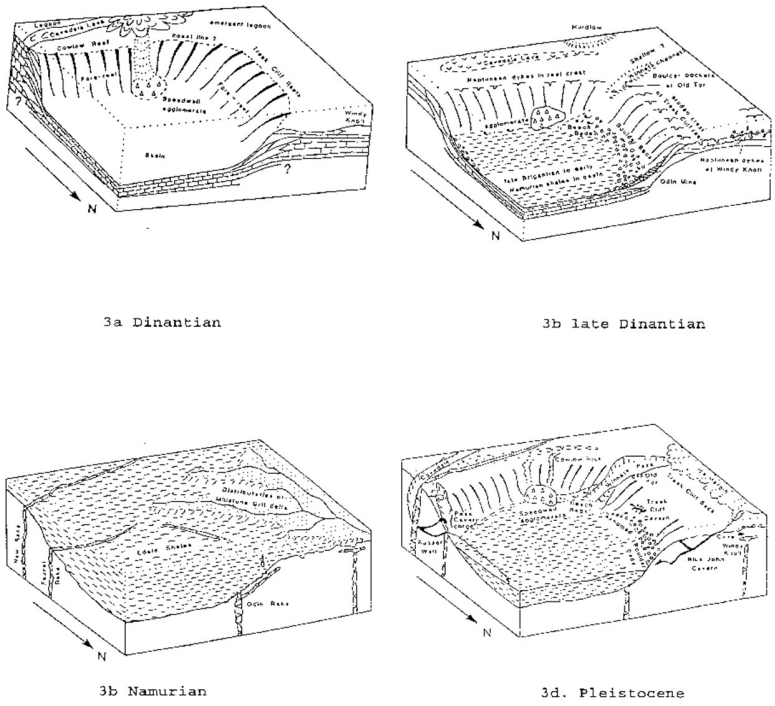


Figure 3. Paleogeographic reconstructions of the Winnats Pass.

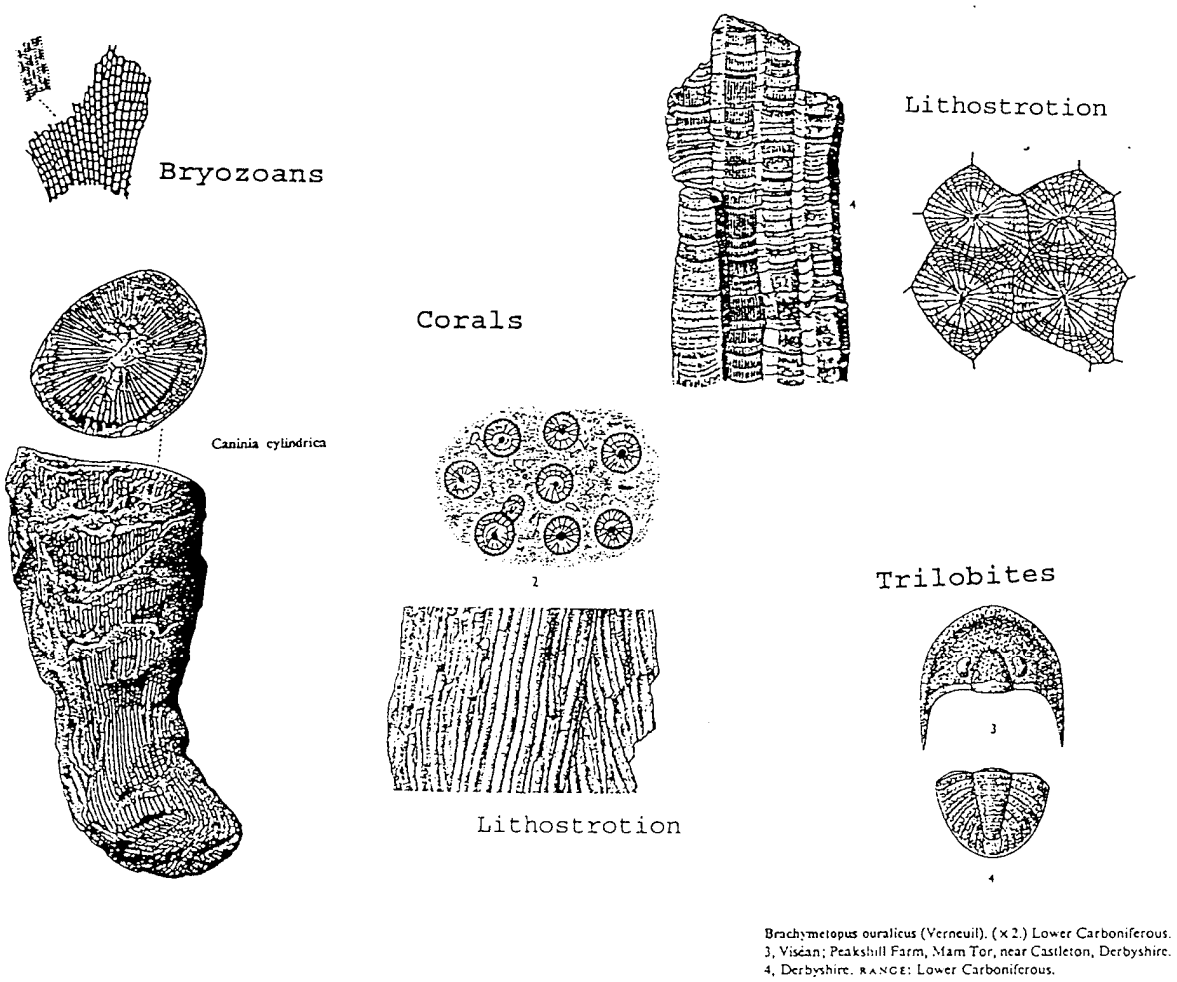


Figure 4a. Typical L. Carboniferous corals and trilobite.

Hillside beneath Treak Cliff (SE 137 830 to 135 833)

Limestones along this slope of the palaeoslope to the shelf are of two types, firstly the in- situ apron reef fossils in limestones with a steep dip (their original attitude) and secondly a boulder facies of material that originated at the top of the slope. The apron reef was a zone rich in preserved fossils with very abundant crinoids. These are representatives the very extensive biota that formed on the platform margin slope (Figs.2, 3a, 4a & b) . Almost all the crinoid fragments are from the stems. The fronds are rarely preserved but are needed for identification. Other fossils present are corals and brachiopods. The boulder material can be found irregularly developed along the slope and contains material with irregular orientations and fossils found in shallow water. In one location, close to the path to Treak Cliff Cavern, just above the road, boulders containing brachiopod and trilobite fragments can be found.

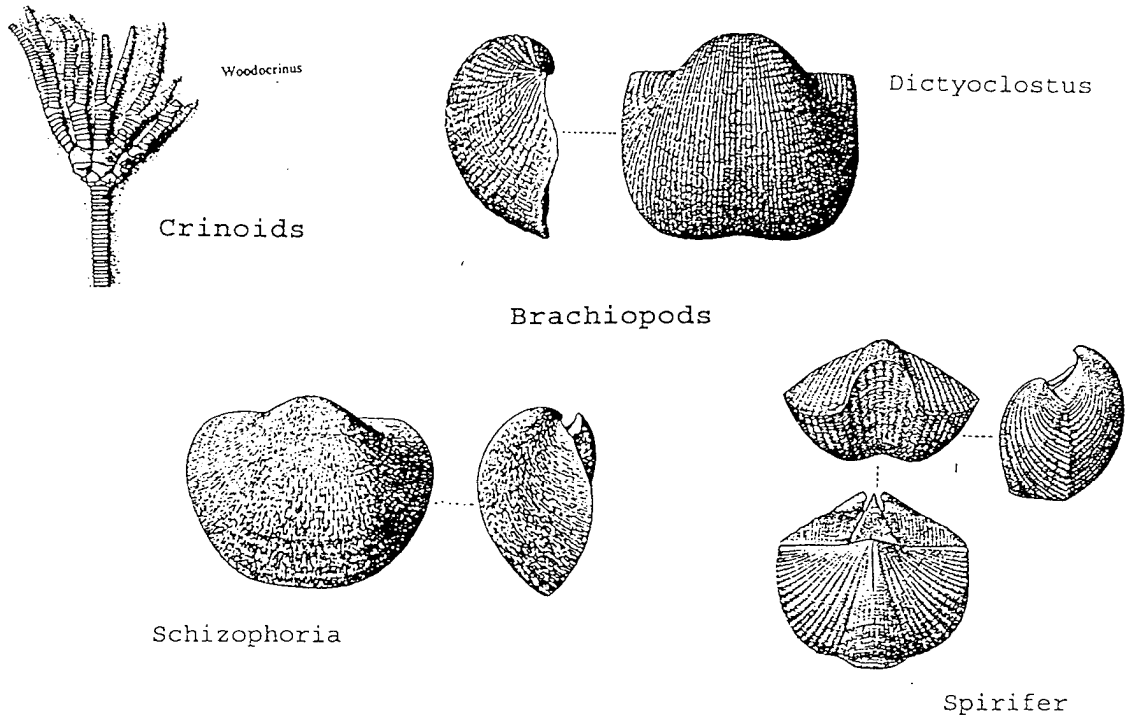


Figure 4b. Typical L. Carboniferous crinoids and Brachiopods.

Geopetals (SE 134 833)

In a gully 300m NW of Treak Cliff cavern (to the north of the track from the Treak Cliff to Blue John Caverns and close to the WW2 bomb crater) there is an outcrop of in situ fossiliferous limestone. In these dipping (slope) limestones there are pores which were partially filled with sediment before being infilled with calcite cement. These are geopetals and can be used as ancient 'spirit levels'. Up the slope from this locality there are several blocks of fossiliferous boulder material and at the top of Treak Cliff, well developed algal stromatolites are found (Fig. 2).

Odin Mine (SE 134 835)

The Peak District has a long history of mining, historically for lead and more recently for barite ($BaSO_4$) and fluorite (CaF_2). Most of the mineralisation occurs in, or adjacent to veins in the limestones (Fig. 6) . The Odin Mine worked the Odin Vein which was the most northerly of the Peak District veins. You can see the east-west trend of the long abandoned vein workings. Minerals present in the vein were galena, PbS, barite, $BaS0_4$, fluorite, CaF_2, calcite, $CaC0_3$ and quartz, $Si0_2$. Although now not abundant, these can still be found on the vein walls and the mine dumps. The old 'ore' crushing wheel is found just below the road.

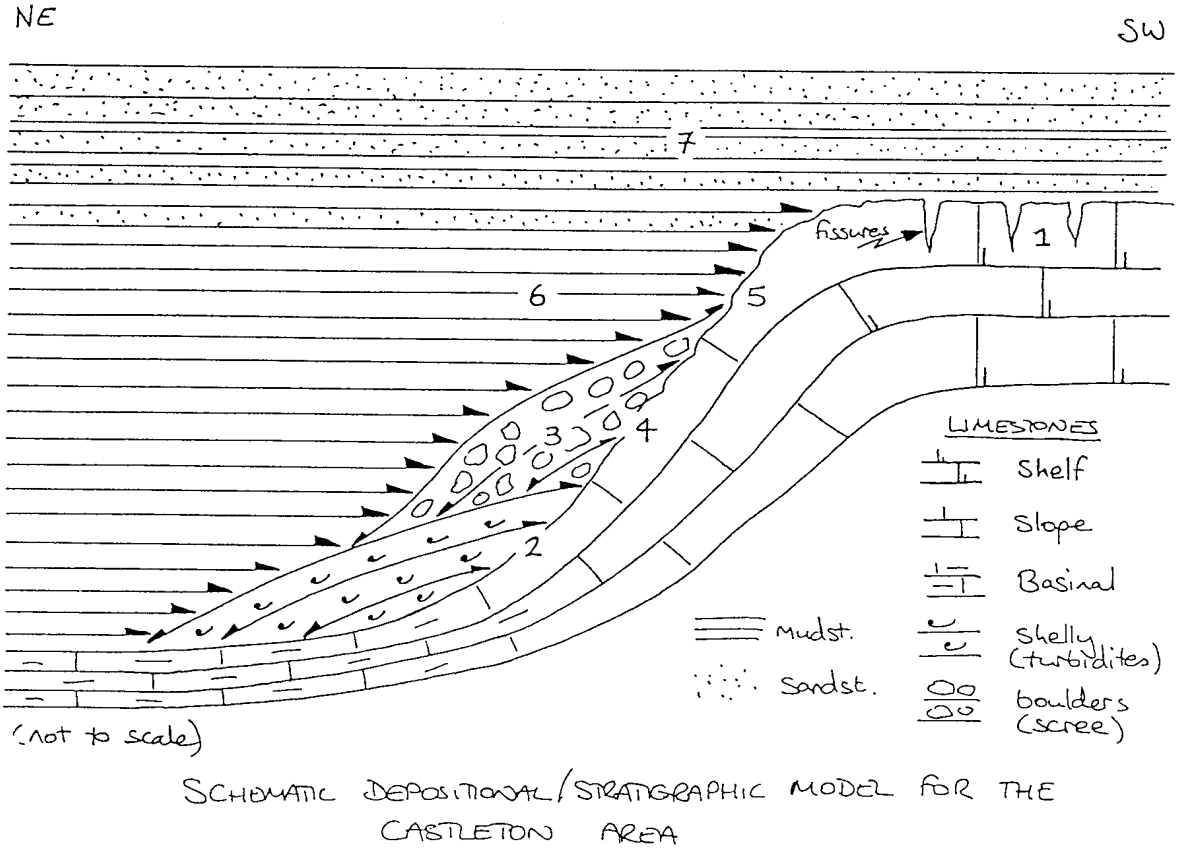


Figure 5. Schematic relationship between the Dinantian and Namurian rocks, looking east. Scale about 2 km horizontally, 3-fold vertical exaggeration.

Current mining in the Castleton area is taking place on Dirtlow Rake, another large vein two miles to the Southeast. Here mining of fluorite and barite is largely confined to a large replacement orebody on the margin of the vein. The mineralisation in the Peak District Orefield formed from hot (100-180°C), saline fluids carrying fluorine, lead and barium rising from depth in the basins surrounding the Peak District limestones. These fluids used faults as channelways and, when they encountered the carbonate rocks, precipitated minerals in the faults to become the veins we see today. These fluids also migrated up the permeable boulder-rich margins of the old carbonate platform, beneath the impervious shales and this is how the fluorite (bluejohn) formed in the Treak Cliff cavern (Fig 7). Later, hydrocarbons (oil) used the same pathways and you can find oil in fractures in the fluorite as well as at Windy Knoll.

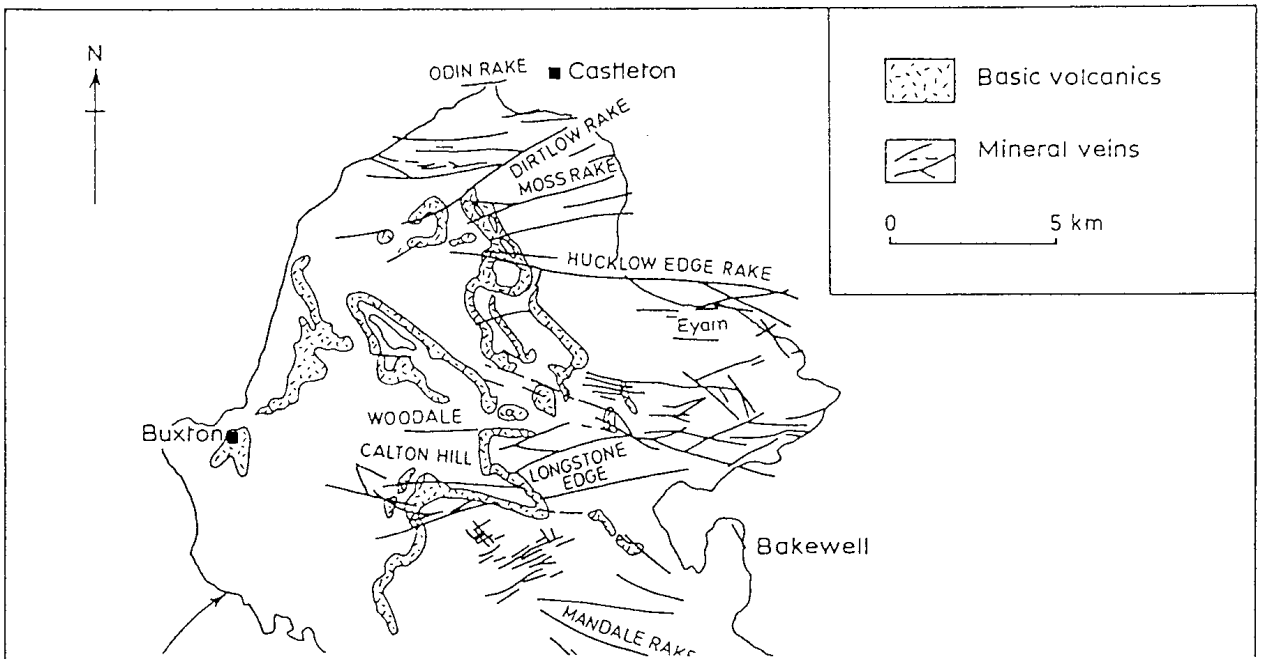


Figure 6. Mineralisation in the Peak District Orefield.

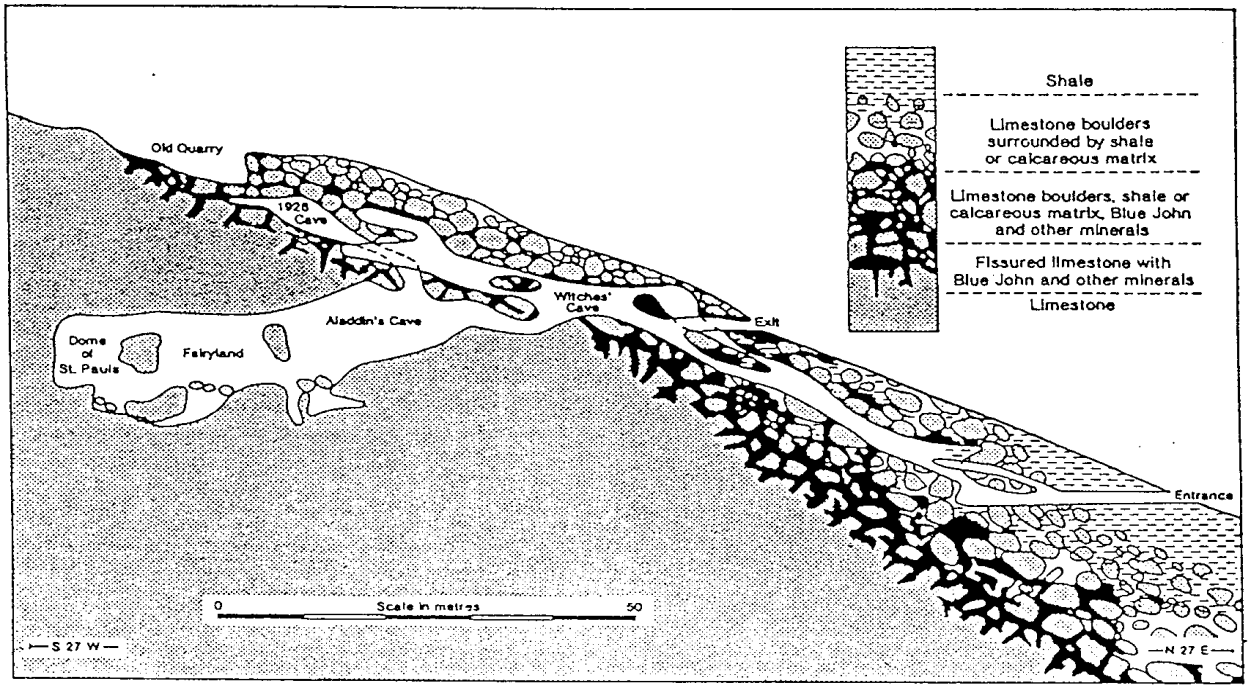


Figure 7. Section through Treak Cliff showing cavern and mineralization.

Mam Tor landslip (SE 132 834)

One of the most striking features of the area is the Mam Tor landslip (Fig. 1 and Figs. 8 & 9). The original cause of the landslip (and others nearby) may be destablization of the slopes following burning of original forest by Mesolithic people. The Mam Tor slip originally formed about 3,600 years ago, although others may be 1000 years older. It is developed within relatively incompetent black mudstones (the Edale Shales) which are overlain by the sandier Mam Tor beds (Fig. 6). It is an excellent place to observe many types of ground movement.

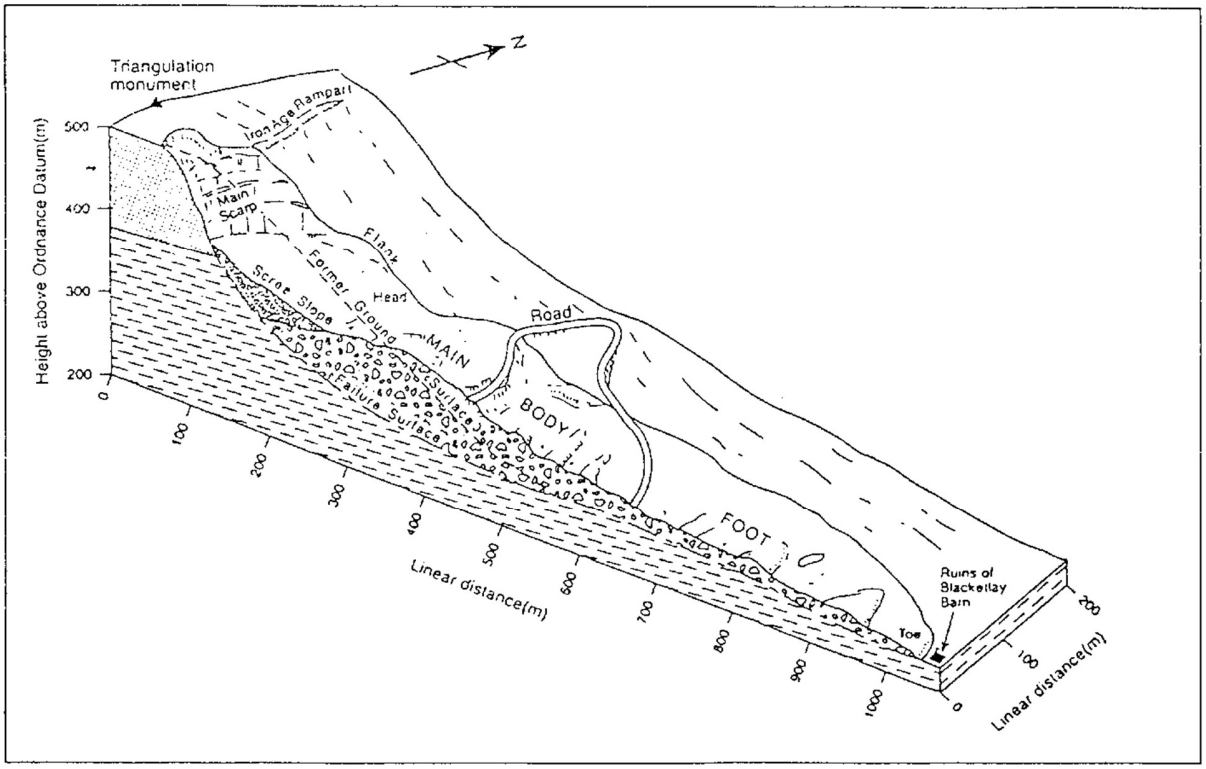


Figure 8. Block diagram to show the structure of Mam tor landslip.

The landslip is moving along a basal fault, about 20 m below the present land surface. Smaller, nested curved slip surface are also developed within the slip, and are best seen where they disrupt the highway. The landslip is still active, moving at up to 50 cm per year, and rainfall seepage into the faults at the base and within the landslip can promote further movement. Stabilization would be very expensive, requiring an extensive drainage system.

After a continuous battle over many decades, the road was finally closed in 1979 owing to the high cost of maintenance. Towards the top of the old road we can see the results of endeavours to keep the road open. The interlayering of tarmac and hard core are witness to several movements on the slip. Interestingly, in places this arrangement mimics sedimentation in natural sedimentary basins formed where subsidence results from vertical fault movements.

The structural geology group at Manchester have been monitoring movements on the landslip annually since 1995 (Fig.8), and this has allowed the pattern of strains within the slip to be defined. Recently, a high-resolution, automatic monitoring station is used to log creep movements at 3 hour intervals, and to correlate them with rainfall and groundwater level

measured in two boreholes. 3 years of monitoring in this way has shown that creep is strongly focused into winter months, not so much because rainfall is focused into the winter months but because vigorous growth of vegetation in the summer recycles about a third of the rainfall back into the atmosphere by evapotranspiration, inhibiting the buildup of groundwater pressure at the base of the slip during the summer. However, exceptionally persistent and heavy rainfall, such as occurred in Jun/July 2007, overwhelmed the capacity of plants to recycle water, and led to an unusual period of summer creep.

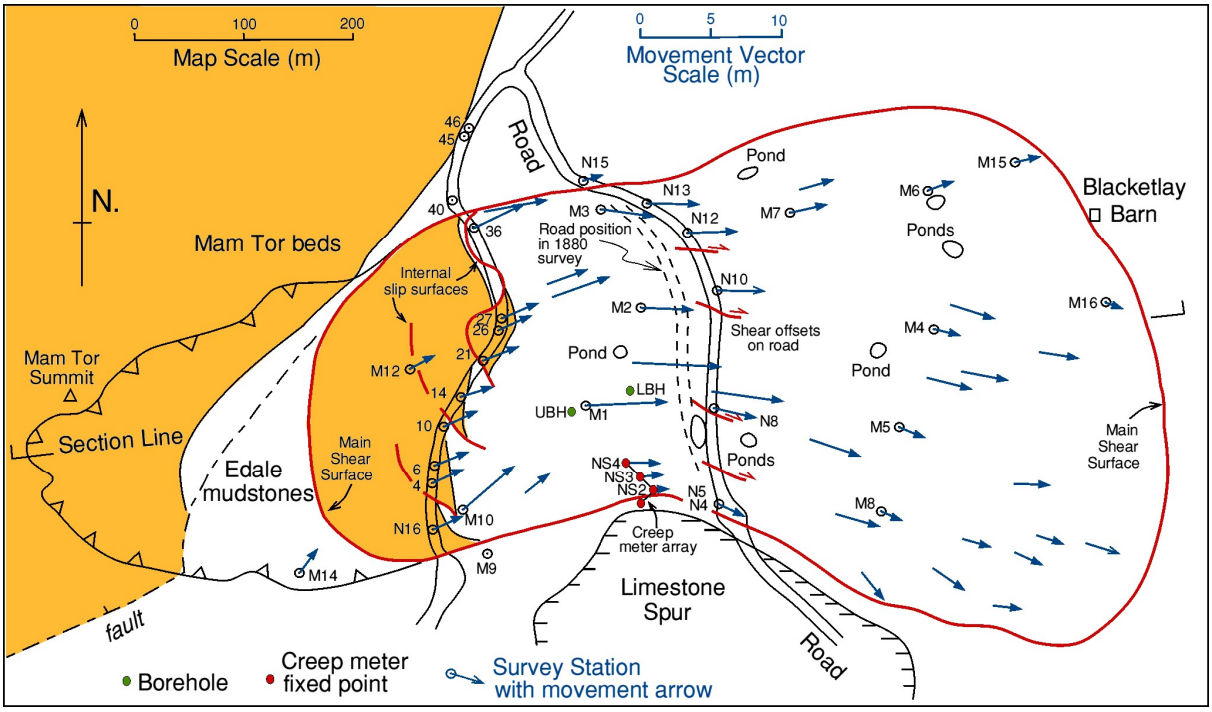


Figure 9. Map showing the outline of the Mam Tor slipped block, and measured movement vectors at stations between 1996 and 2008. Movement directions follow precisely the same pattern year on year. Note how the lower (eastern) arm of the highway has displaced downhill by 40m since 1880.

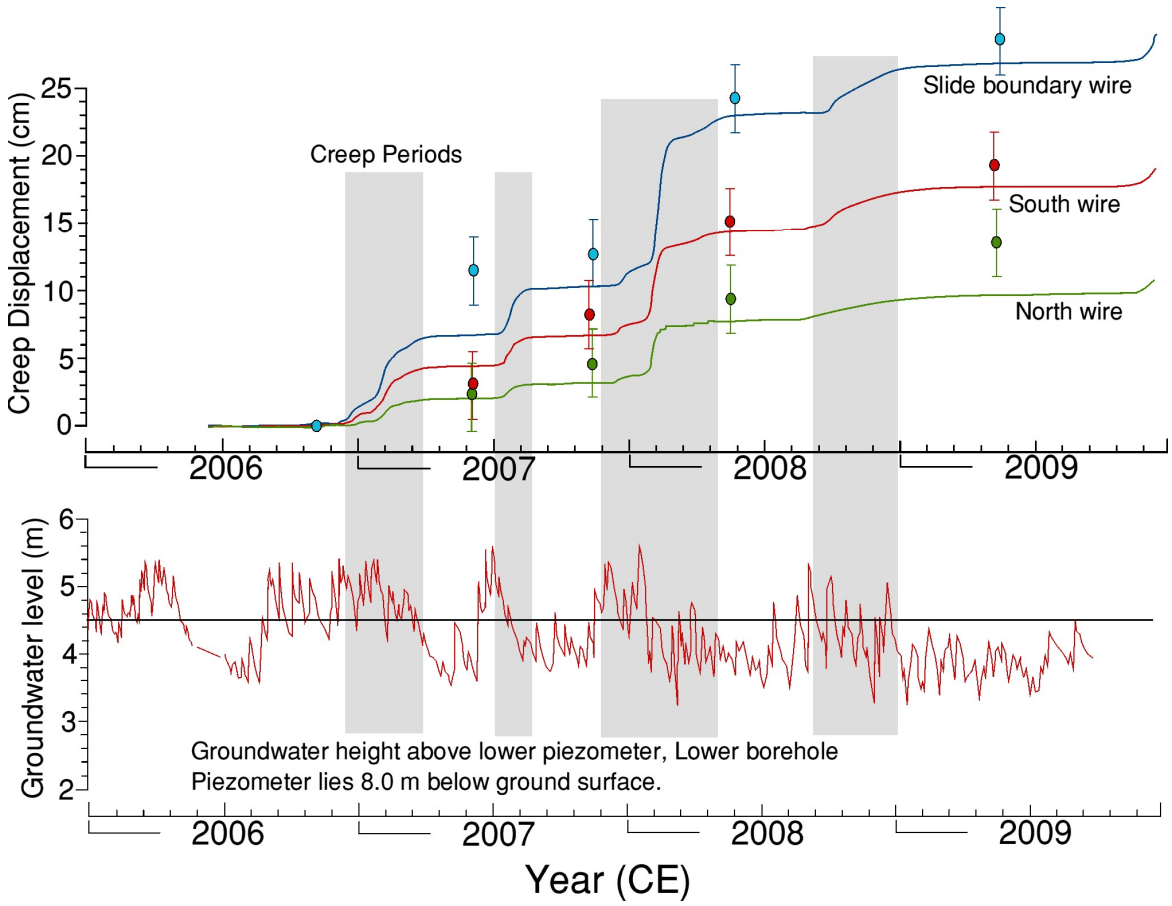


Figure 10. Time-series data obtained from the three creep meters (above) and the groundwater pressure (metres) obtained from the lower piezometer in borehole LBH (lower graph). The piezometer lies 8.0 m below the present ground surface. The horizontal line at 4.5 m is arbitrarily drawn as a reference line to help contrast the periods of relatively high and low groundwater pressure. The short term fluctuations correspond to individual rainfall events. The vertical grey bars correspond to principal creep periods, that occur when groundwater levels are high. Creep displacements are shown parallel to the wires, and the plotted points are measurements made by electronic distance measurements at the ends of each wire.

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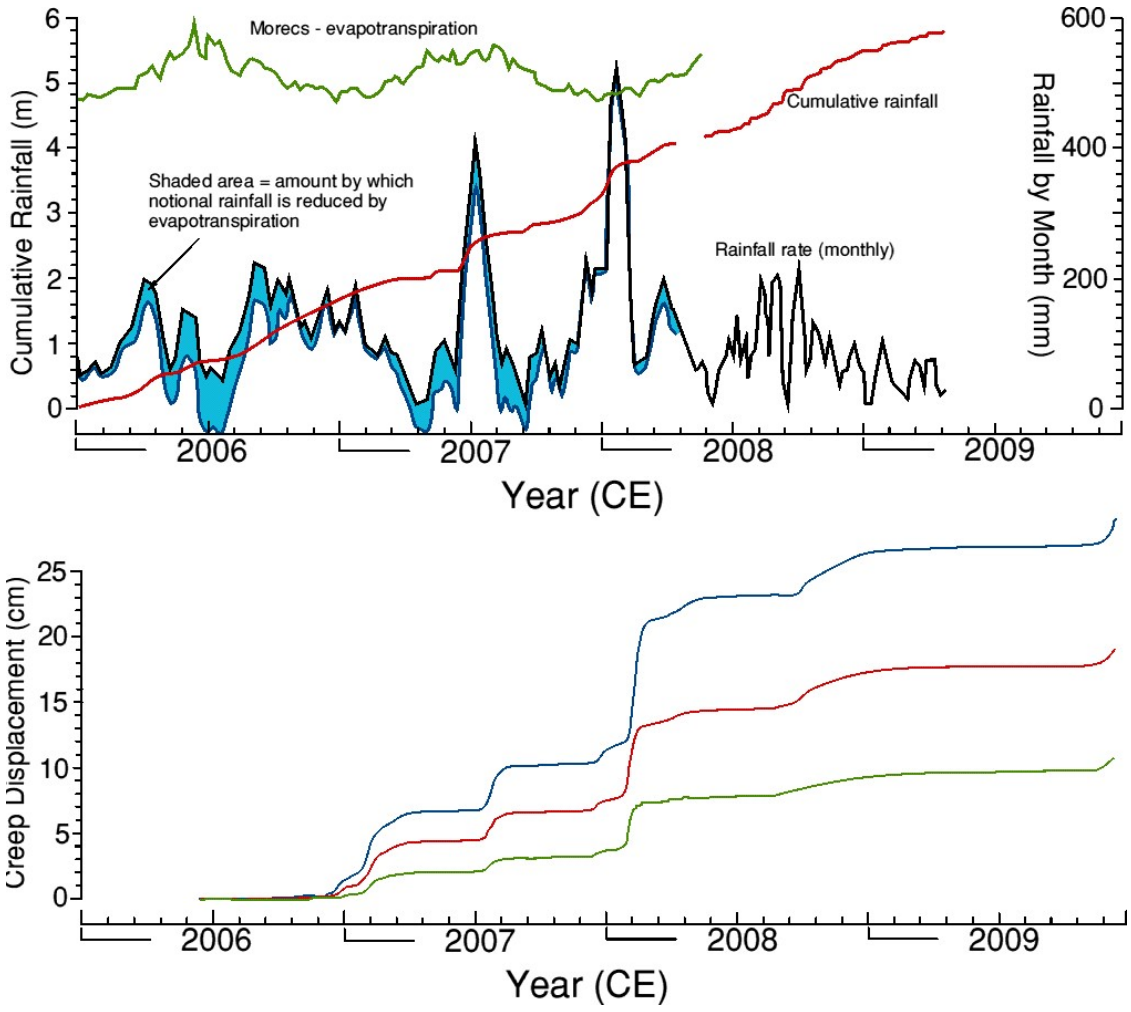


Figure 11. The relationship between creep (lower graph) and rainfall (upper graph) at Mam Tor. Cumulative rainfall and monthly rate of rainfall are shown. There is a strong correlation of creep with the marked rainfall events in summer 2007 and winter 207-08. The uppermost graph shows the notional loss of infiltrating rainfall due to evapotranspiration by plants, and this amount has been subtracted from the total rainfall rate graph for the corresponding period (shaded interval). Evapotranspiration significantly reduces the amount of penetrating rainfall in the summer months, and this helps explain the polarization of creep into the winter months. However, as the summer 2007 shows, this effect can be swamped by extreme rainfall.

Bog iron ore (SE 136 837)

One interesting consequence of the landslip is that we can see a mineral deposit actually forming today. The Edale shales are rich in iron in the form of pyrite, FeS_2. Water is percolating into the shales at the top of the landslip, possibly using the various fault surfaces. This water is reacting with the iron in the shales and becomes acidified, leaching out iron. While underground, the iron stays in solution. After migrating through the landslip these waters reappear as springs in the lower ground, at and below the road. Here the waters

emerge and are 'oxidised' causing immediate precipitation of the iron oxides and hydroxides. We can see the hard iron-rich mineral crusts around the springs. These deposits are of a type called 'Bog-Iron' ore. Although not economically important, such ores have been worked on a small scale the world over.

Mam Tor (SE 130 837)

In the base of Mam Tor (Fig. 1) we can see both the Edale Shales and the Mam Tor beds. The Edale shales are pyritic, organic-rich marine mudstones. These were formed in deep water between the carbonate shelf and advancing deltas to the north. They represent terriginous material brought down by rivers and washed far out into the basin, away from the delta front. Rare *pectens* shells can be found.

The pyrite, FeS_2, in the mudstones was originally formed by sulphate reducing bacteria in the sediment pore waters. However, at the surface, when exposed to air and oxygenated, the pyrite is unstable and converts to iron oxides (rust), one product being the production of sulphuric acid by the reaction:

$$4FeS_2 + 15O_2 + 8H_2O = 2Fe_2O_3 + 8H_2SO_4$$

The acid formed helps keep iron in solution, an important part of the process of formation of the bog iron ore at the base of the landslip. This same process of acid formation causes much damage around metal and coal mining areas, resulting in so called 'acid mine drainage'. At Mam Tor we can also see the fate of some of the sulphuric acid. Here it reacts with carbonate minerals to form the mineral gypsum, $CaS0_4.2H_20$ by the reaction:

$$CaCO_3 + H_2SO_4 + H_2O = CaS0_4.2H_2O + CO_2$$

The summit of Mam Tor is made of interbedded mudstones and sandstones. The sandstones are medium grained and at the bases of the units, clear indicators of current activity and direction can be seen. The current had scoured the muds and then sand was deposited, preserving the scours which now form casts at the base of the sandstone units. These are termed 'flute' and 'groove' casts and can indicate the direction of sediment transport. These sandstones herald the onset of true deltaic sedimentation during the Upper Carboniferous (Fig. 3c). The Castleton area was covered by deltas developing from the north. These first signs are in fact turbidites, that are gravity flows of material forming at the front apron of the delta. Plant debris and very rare fish are found in the sandstones.

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